

CISCO Packet Tracer Lab 1 Report

Basic Network Setup and Connectivity

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Lab Objective

The objective of this lab was to build a basic LAN in Cisco Packet Tracer by connecting a router, switch, and PCs. We configured the router's interface with an IP address, set up DHCP, and assigned static IP addresses to each PC. The lab focuses on verifying network configurations using ipconfig and show ip interface brief commands, then testing connectivity between all devices using ping to demonstrate successful communication.

Tasks:

1. Identify and label each device in the given network diagram.
2. Assign static IP addresses to computers and configure DHCP on the router.
3. Use the ping command to verify connectivity.
4. Utilize ipconfig to confirm network settings.

Equipment and Resources

Hardware:

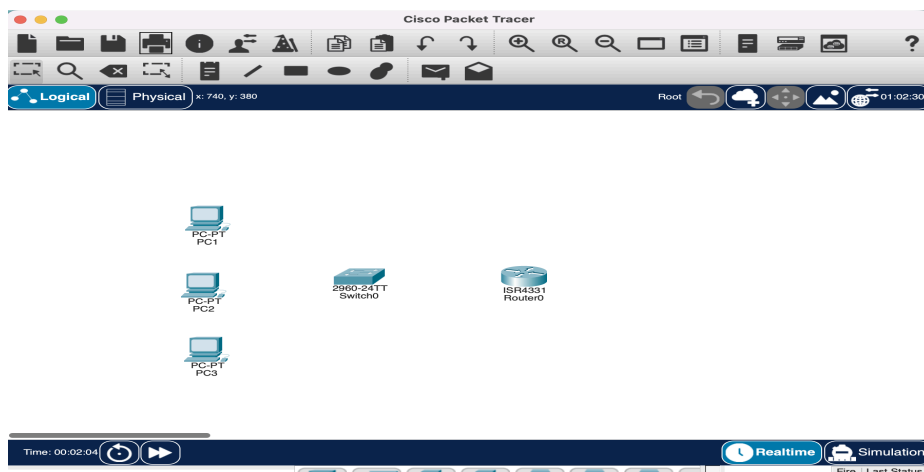
- 1x Cisco ISR4331 Router
- 1x Cisco 2960-24TT Switch
- 3x PC End Devices
- 4x Copper Straight-Through Cables

Software:

- Cisco Packet Tracer (Network Simulation Software)

Lab Steps

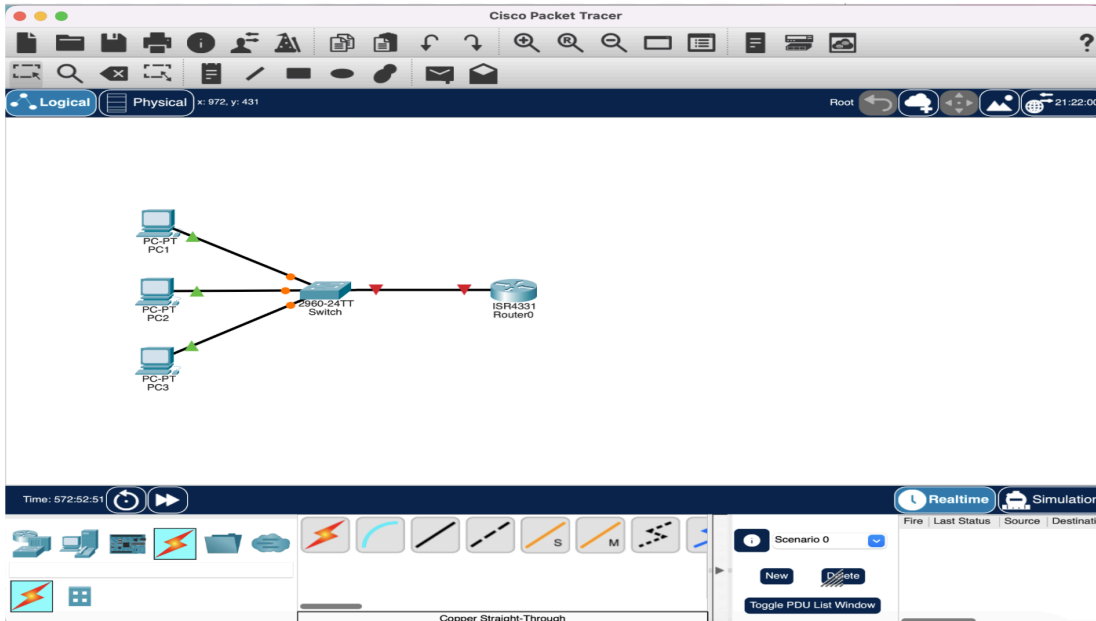
Step 1: Initial Network Topology



First, we added one ISR4331 router, one 2960-24TT switch, and three PC end devices. They were arranged in a simple layout

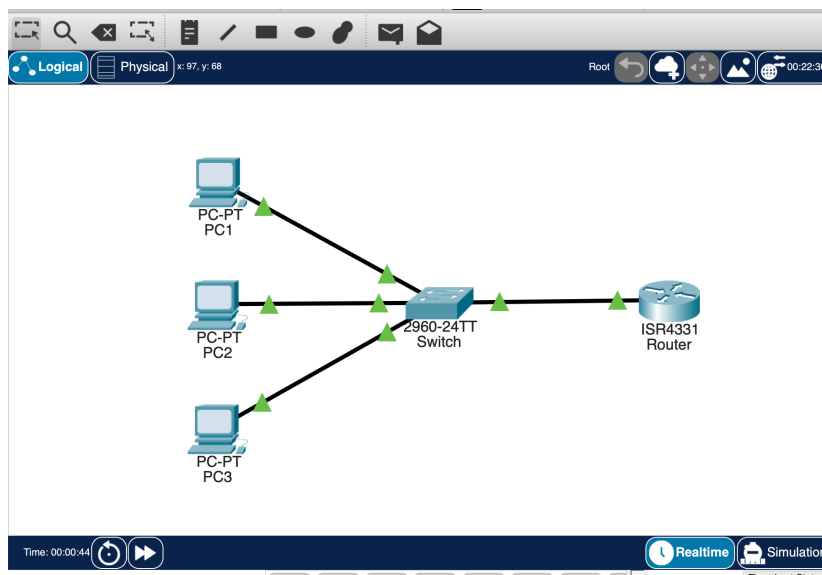
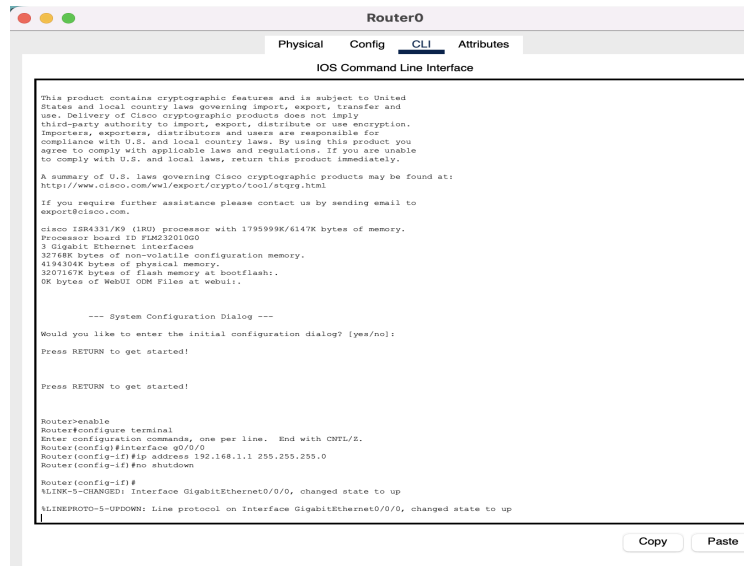
where all PCs connect to the switch, and the switch connects to the router. This created a basic LAN structure for us.

Step 2: Network with All Connections



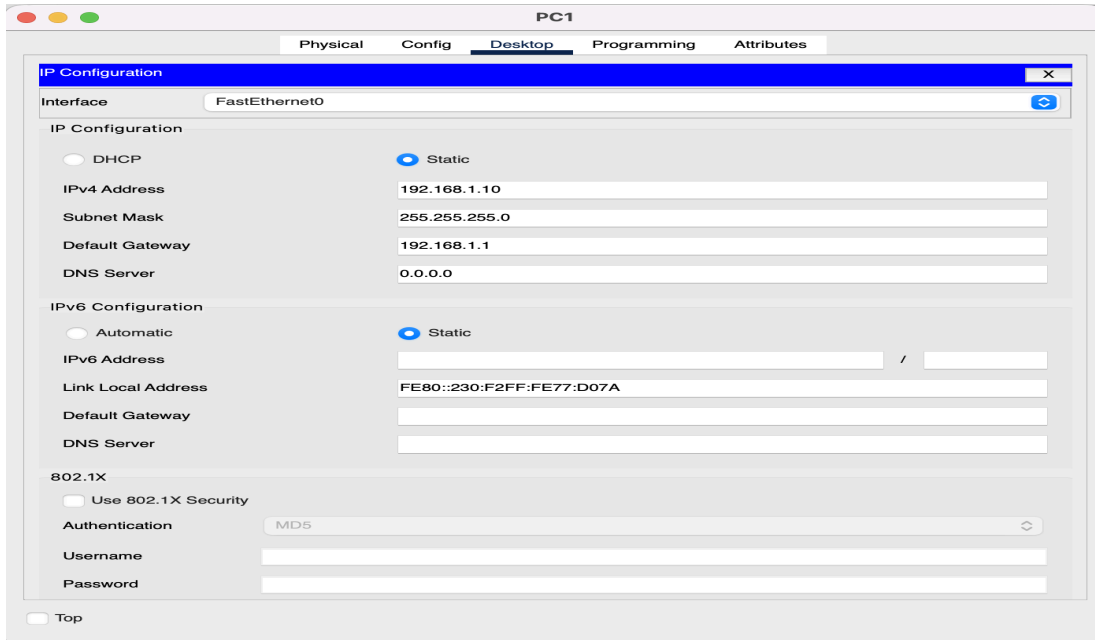
We connected all the devices using Copper Straight-Through cables. PC1, PC2, and PC3 were connected to the switch on ports FastEthernet0/2, 0/3, and 0/4. Then the switch was connected to the router's GigabitEthernet0/0/0 port. Once everything was plugged in correctly, the connection indicators didn't turn green, which meant the physical layer couldn't working properly without configuring the router.

Step 3: Router CLI Configuration



Next, we accessed the router through the CLI to set up the network settings. We gave the GigabitEthernet0/0/0 interface the IP address 192.168.1.1 with a subnet mask of 255.255.255.0. We had to use the 'no shutdown' command to turn the interface on. After that we created a DHCP pool for the 192.168.1.0/24 network and set the default router to 192.168.1.1 so devices could get IP addresses automatically. Configuring the router turned the lights green

Step 4: PCs IP Configuration

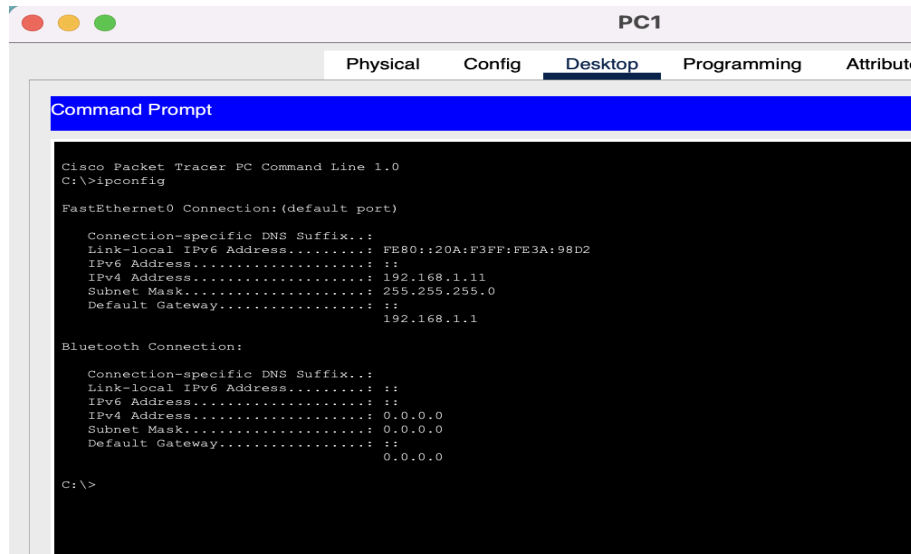


We manually configured PC1 with a static IP address by going to the Desktop tab and clicking on IP Configuration. We selected Static instead of DHCP and entered 192.168.1.10 for the IP address, 255.255.255.0 for the subnet mask, and 192.168.1.1 for the default gateway. This way PC1 always has the same IP address. The same process was repeated for PC2 (192.168.1.11) and PC3 (192.168.1.12)

PC IP Table:

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.10	255.255.255.0	192.168.1.1
PC2	192.168.1.11	255.255.255.0	192.168.1.1
PC3	192.168.1.12	255.255.255.0	192.168.1.1

Step 5: ipconfig Output on PC



The screenshot shows a window titled "PC1" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the output of the "ipconfig" command, indicating the IPv4 address is 192.168.1.11, the subnet mask is 255.255.255.0, and the default gateway is 192.168.1.1.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::20A:F3FF:FE3A:98D2
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.11
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 192.168.1.1

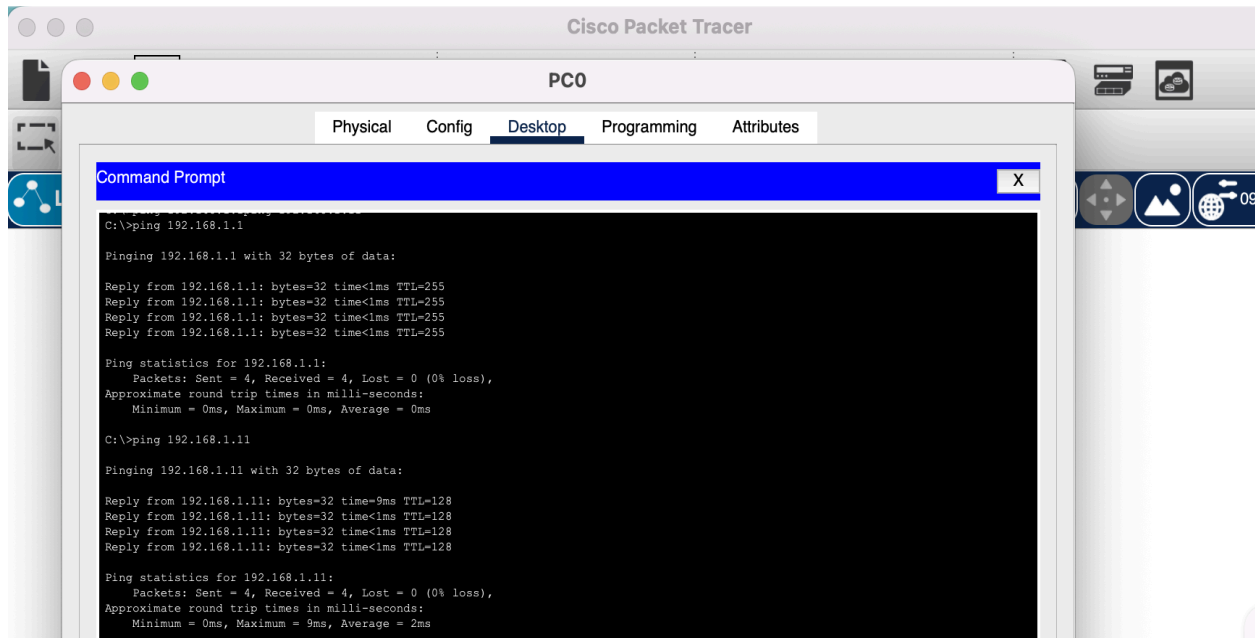
Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>
```

To make sure the PCs were configured correctly, we opened the Command Prompt from the Desktop tab and ran the ipconfig command for all 3 PCs. For example when we tested PC1 the output showed us that the IPv4 address was 192.168.1.11, the subnet mask was 255.255.255.0, and the default gateway was 192.168.1.1, which confirmed everything was set up right.

Step 6: Ping Test Results



For the final test we used the ping command to see if everything was connected. For example, we pinged PC1. We pinged the router at 192.168.1.1 and got back 4 replies with no packet loss. Then we pinged PC1 at 192.168.1.11 and PC3 at 192.168.1.12 and both worked perfectly with 0% packet loss. This proved that all the devices could communicate with each other and the network was working.

Results and Observations

The network was successfully implemented using Cisco ISR4331 as the router, one Cisco 2960-24TT switch, and three PCs. The PCs were connected to the Switch and the Switch to the router to establish a network. A DHCP was configured and the PCs were assigned static IP addresses within the same subnet. The *ipconfig* command confirmed that each PC had the expected IPv4 address. Connectivity testing using the *ping* command demonstrated successful communication between devices. All echo requests returned replies with 0% packet loss, confirming proper configuration between the layers 1, 2, and 3 (physical, switch communication, and IP addressing respectively). The results indicate that the network was correctly configured and fully operational.

Challenges Faced and Solutions:

One of the initial challenges we encountered was the incorrect IP configuration of the end devices. Even a small mismatch could prevent a device from communicating properly. To solve this issue we used the *ipconfig* command to validate settings and confirm the correct IP address, subnet mask, and gateway for each of the PCs.

At a later stage we encountered that devices were unable to communicate despite appearing correctly configured. We decided to approach this problem by using the *ping* command to test the connectivity beginning with the router interface and then between PCs. Additionally, we then checked the cable connection status and confirmed the router interface was active.

A third problem we encountered was the DHCP not correctly assigning the IP Addresses. Although a DHCP pool was created, client devices initially failed to receive an IP address automatically. To solve this we verified the DHCP configuration on the router and restarted the client device to trigger a fresh address request. Once the DHCP service was correctly validated, automatic IP assignment worked as expected.

Lessons Learned:

1. A perfect physical link appears when the switch light is green, but IP configuration flaws can still cause ping failures.
2. The router's inner interface requires proper gateway addresses for computers to communicate with neighboring PCs.
3. *ipconfig* reveals the network settings applied from our configuration policy.
4. *ping* tests confirm a clear round-trip path and help diagnose issues with cables, IP, settings, or firewall blocks.

Real-World Applications:

The skills practiced in this lab can directly apply to real-world networking environments. For example, in hospitals, networks play a role in making sure medical devices along with record systems of the patients can stay connected and reachable throughout. In schools and universities networks manage campus-wide laboratory computers through centralized DHCP services, while administrators regularly test connectivity to ensure reliable communication across the network. In home networks routers are capable of connecting automatically to IP addresses that are assigned while users troubleshoot using *ping* and *ipconfig*. In retail stores cash registers and payment terminals have to be connected to the store's network in order for credit cards to process transactions.

Conclusion:

Overall, we successfully built a basic LAN in Cisco Packet Tracer using the ISR4331 router, 2960-24TT switch, and three PCs. We handled cabling, set the router's GigabitEthernet0/0 interface to 192.168.1.254 with a DHCP pool for the 192.168.1.0/24 network, and configured static IPs as required. After fixing an initial DHCP glitch with *ipconfig* and ping checks, everything connected perfectly—no packet loss on any tests. These steps showed real-world debugging skills, like checking switch lights and gateways first and they will help us greatly when moving forward to more complex setups.